

Aligning Drone Strategy 2.0 With Horizon-EU Cluster 5 Destination 6 – Topic 11 “IAM: Drone Logistics”

European Commission Drone Strategy 2.0 (DS2.0) sets out a holistic blueprint to integrate drones and eVTOLs into European airspace and markets by 2030. It introduces 19 flagship actions spread across ten thematic areas to 1) build a Union-wide drone-services market and 2) strengthen EU civil-security-defence capabilities. Five of these flagships—Actions 1, 5, 13, 17, 18—are particularly relevant to the Horizon Europe Cluster 5 Destination 6 Topic 11 call on “Innovative Air Mobility (IAM): Drone logistics for ports, harbours and critical urban infrastructure”.

Below is an in-depth examination of the strategy, the specified flagship actions, and practical guidance on how to weave them into a winning proposal that prioritises safety, security, connectivity, traffic-management and business viability for port-centric drone logistics.

Overview

Drone Strategy 2.0 (adopted 29 Nov 2022) charts the path for an EU drone ecosystem that is safe, green, digital and economically robust by 2030^{[1] [2]}. It envisions:

- Seamless integration of manned and unmanned traffic via U-space and future Integrated CNS services^{[3] [4]}.
- A €14.5 billion EU drone-services market and 145,000 jobs by 2030^{[5] [4]}.
- Dual-use civil–defence innovation synergies to underpin strategic autonomy^{[6] [7]}.

Port, harbour and critical-infrastructure logistics feature prominently as early commercial use-cases thanks to their fenced perimeters, predictable flight corridors, and high value-per-kg cargo (spares, medical supplies, time-critical documents)^{[8] [9]}.

The Five Flagship Actions

Flagship Action	DS2.0 Theme	Core Deliverable	Relevance to Port/Harbour Logistics
1 – Amend SERA & ATM/ANS	Airspace integration	Integrate drones/eVTOL into Standardised European Rules of the Air (SERA) and ATM/ANS Regulation for mixed-traffic safety ^[1]	Enables BVLOS corridors that traverse CTRs and naval approaches without bespoke exemptions, reducing ops cost and approval lead-time ^[10]

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5 – Vertiport Rules	Innovative Air Mobility	EASA to issue binding rules for vertiport design & ops under Basic Regulation, building on Prototype Technical Design Specifications ^[11] ^[12]	Standardised mini-vertiports on quay walls, container decks or offshore platforms, with clear firefighting, obstacle-free volume and noise contours, accelerating infrastructure permitting in congested docks
13 – Spectrum Plan	Industrial capabilities	EU-wide coordination to guarantee sufficient, interference-protected radio-frequency spectrum for C2 links and U-space ^[2]	Protects redundant SATCOM/5G links amid heavy maritime radar traffic; ensures deterministic latency for Automatic Dependent Surveillance–Broadcast (ADS-B) in busy vessel traffic service (VTS) zones
17 – Counter-Drone Package (C-UAS)	Security & resilience	Legislative/technical toolkit for detection, identification, neutralisation of rogue drones; handbook for critical infrastructure ^[2] ^[13]	Safeguards oil terminals, LNG tanks, port cranes against malicious UAV; clarifies legal use of RF jammers, spoofers, intercept drones, kinetic defeat systems within port estates ^[14] ^[15] ^[16]
18 – Aviation Security Amendments	Security & resilience	Amend current aviation-security rules so airports & authorities bolster resilience to drone risks ^[17]	Extends mandatory drone-incident response plans to multi-modal hubs (sea-air-rail) and sets training standards for harbour masters, coast guards, police, and private security to manage drone incursions

SERA, U-space and the Port Context

Why SERA Matters

SERA sets uniform flight rules across EU airspace. Flagship 1's amendments will:

- Embed detect-and-avoid (DAA) alternatives to “see-and-avoid” for unmanned missions^[1].
- Define harmonised contingency procedures for drones losing C2 within a port's CTR or maritime traffic lane, preventing “grey zones” in jurisdiction between ANSPs and VTS.

U-Space Layers Beside VTS

Modern ports run Vessel Traffic Services. Integrating U-space service providers (USSPs) with VTS enables:

- Real-time geofencing of exclusion zones (e.g., around LNG carriers) broadcast to drones.
- Automated de-confliction between rotorcraft, cranes, tall-mast vessels and drone corridors.
- Shared spectrum (Flagship 13) for maritime AIS and U-space C-network to avoid RF clashes^[2].

Mapping DS2.0 Flagships to Horizon Call Objectives

1. Safety & Security of Critical Infrastructure

- **Action 17/18:** Embed a layered C-UAS architecture at quay, perimeter and airspace levels. Demonstrate integration of radar-RF-EO sensors with Port-Community Systems (PCS) for automatic threat classification and neutralisation workflows compliant with the forthcoming EU C-UAS package^{[15] [16]}.
- **Action 1 (SERA):** Validate emergency flight-termination/alternate-landing protocols using designated port heli-ramp surfaces or floating vertiports.

2. Connectivity & Traffic-Management Pivot

- **Action 13 (Spectrum):** Prototype dual-redundant C2 using 5G NTN (Non-Terrestrial Networks) plus protected aviation C-band, ensuring 99.9% link availability over reflective water surfaces.
- **Action 5 (Vertiports):** Build modular, ISO-containers-based vertiport pads with automated battery swap stations. Show compatibility with EASA design specs for fire safety, obstacle-free funnel, and 65 dB(A) noise contour^{[11] [12]}.

3. Economic & Business Outcomes

- **Action 5/13:** Lower infrastructure CAPEX via standardised vertiport templates and future-proof spectrum rights, improving ROI for logistics operators.
- **Action 1:** Uniform SERA revisions cut permitting costs and facilitate cross-border coastal flight corridors for pan-European liner services (e.g., Rotterdam-Hamburg-Oslo).

Proposal Blueprint for Topic 11

Strategic Vision

Establish a “**Port-U-Link Living Lab**”—a replicable, multi-site demonstration in three EU seaports (core TEN-T network) showcasing safe, secure, connected drone logistics for critical-infrastructure supply chains and emergency response.

Work-Package (WP) Structure

WP	Title	Flagship Alignment	Key Outputs
WP1	Regulatory Harmonisation & Safety Case	1,13	SERA-compliant Specific Operations Risk Assessment (SORA) + draft vertiport certification dossiers
WP2	U-Space-VTS Integration Layer	1,13	API bridge, joint ATM-VTS dashboard, latency testing under multi-sim fault injection
WP3	Port Vertiport Prototyping & Spectrum Trials	5,13	Deploy two containerised vertiports per port; validate C2 link resilience with spectrum analytics

WP	Title	Flagship Alignment	Key Outputs
WP4	Secure BVLOS Logistics Operations	1,5	2,000 BVLOS flights moving urgent spares, medical kits, customs documents; track KPIs (on-time, emissions, cost)
WP5	Counter-UAS & Resilience	17,18	Multi-sensor C-UAS mesh; red-team penetration tests; updated Port Security Plan
WP6	Societal Acceptance & Business Models	5,17	Noise-signature tests, stakeholder co-creation labs, LCOE vs. truck/bike baseline models
WP7	Replication Toolkit & Policy Roadmap	1-18	Open-access blueprint, training modules, investment guide for EU/third-country ports

Technical Enablers

- **Detect-and-Avoid:** On-board ADS-B In/Out & optical flow sensors feeding cooperative separation services^[4].
- **Redundancy:** Hybrid SATCOM-5G SA links leveraging Flagship 13 spectrum plan.
- **U-Space Services:** Strategic de-confliction, tactical conflict resolution, conformance monitoring and emergency management^[1].
- **C-UAS:** Passive RF detection, radar-EO fusion, AI-assisted classification; DroneGun Mk4/DroneCannon Mk2 for controlled defeat^{[15] [16]}.
- **Noise Abatement:** Low-RPM seven-blade rotors; dynamic routing to avoid residential waterfronts^{[18] [11]}.

KPIs & Targets

Indicator	Baseline	Target (Project End)
Incident-free flight hours	0	1,200 h with zero ground-risk incidents ^[4]
Average door-to-door lead-time (lab samples)	150 min	≤30 min (-80%) ^[19]
Cost per delivery (≤5 kg, 10 km)	€55	€12 (after 1-operator-5-drones ratio) ^{[19] [9]}
CO ₂ -eq reduction vs. diesel van	0%	≥85% ^{[8] [9]}
Successful rogue drone interdictions in red-team drills	n/a	≥95% detection, ≥90% neutralisation ^{[14] [15]}

Risk Management

Risk	Mitigation
SERA amendment timelines slip beyond 2025	Build ops under "Specific" category with enhanced SORA; align with EC/EASA transition guidance ^{[1] [10]}
Spectrum congestion in busy ports	Early coordination with national spectrum regulators; deploy dynamic spectrum access algorithms, protected C-band fallback ^[2]

Risk	Mitigation
Public acceptance of noise/visual intrusion	Deploy low-noise rotors, restricted flight windows, real-time community noise map apps ^[18]
Cyber-hacks on U-space-VTS API	Zero-trust architecture, penetration testing, AES-256 C2 encryption, bug-bounty programme ^[9] ^[20]

Exploitation & Business Models

1. **Infrastructure-as-a-Service (IaaS)**: Port authority licences vertiport decks and U-space nodes to multiple drone-logistics providers (open access aligns with Flagship 5).
2. **Logistics-as-a-Service (Laas)**: Consortium spin-off operates drone fleet for ship-to-shore and intra-terminal routes; subscription model for shipping lines.
3. **Security-as-a-Service (SECaaS)**: C-UAS sensor-network offered to neighbouring energy plants or coastal airports via monthly fee, monetising Flagship 17 compliance.

Projected IRR: 21% by Year 5 on €15 million combined CAPEX (vertiports, USSP licences, drones), assuming 18,000 annual flights per port at €25 revenue each^[8] ^[9].

Contribution to EU Green & Digital Objectives

- **Green Deal**: Replaces diesel vans for “last nautical mile” transfers; anticipated savings of 490 t CO₂-eq per port annually^[8].
- **Digital Decade**: Demonstrates 5G NTN performance for aviation; enhances European strategic telecom autonomy^[2].
- **Strategic Compass**: Validates dual-use C-UAS shield at critical maritime nodes, supporting EU resilience aims^[7] ^[13].

Implementation Timeline (Gantt Snapshot)

Year	Q1	Q2	Q3	Q4
2025	Grant prep, SERA liaison, spectrum licence	Vertiport build start	U-space-VTS interface, first tethered flights	BVLOS trials, C-UAS install
2026	Scale to 5 drones/operator, 500 flights	Manned-eVTOL test (Phase In)	Noise/social studies	Business model stress-test
2027	Cross-border corridor demo	Full regulatory hand-off to ANSP	Replication workshops EU/ICAO	Commercial go-live

Policy Recommendations

1. **Fast-Track Vertiport Permits** within TEN-T nodes under a single-window maritime-aviation authority to implement Flagship 5 quickly.
2. **Spectrum Reservation**: Allocate harmonised 5030–5091 MHz aviation C2 band with maritime coexistence guidelines (Flagship 13).

3. **Hybrid Safety Case** templates that integrate SERA (air), ISPS Code (sea) and port-state control audits, supporting Flagship 1/18 mutual recognition.
4. **EU Funding Synergy:** Align CEF Transport calls with Horizon Pilot-scale implementations for follow-on deployment of port vertiports.

Conclusion

Drone Strategy 2.0 offers a concrete regulatory and technological scaffold for port and harbour drone logistics. By explicitly targeting Flagship Actions 1, 5, 13, 17, 18, proposers can deliver a demonstrator that not only satisfies Horizon Europe Topic 11 goals but also accelerates EU-wide rollout of safe, secure, spectrum-robust, U-space-enabled logistics services—unlocking both economic gains and critical-infrastructure resilience for the next decade.

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